Structural Re-Parameterization, Spatial Coordinate Encoding, and Cross-Domain Robustness for Real-Time Safety Helmet Detection in Construction Surveillance

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***Abstract*—Automated Personal Protective Equipment (PPE) compliance monitoring via surveillance cameras is essential for maintaining occupational safety on industrial construction sites. However, edge-based surveillance pipelines encounter severe operational bottlenecks, including distant tiny helmets, extreme category imbalance between workers and safety gear, dense physical occlusions, and background clutter. To overcome these limitations, we propose Rep-YOLO11s, an efficient edge detector engineered as a hardware-aware architecture tailored for memory-constrained edge deployment. Our architecture addresses single-batch streaming latency bottlenecks by incorporating structural re-parameterization to fold parallel training branches into single convolutions during deployment, minimizing Memory Access Cost (MAC). To suppress false alarms from ground-level clutter, stem coordinate convolutions provide elevation-aware spatial priors. In addition, bi-level routing attention is integrated into neck stages to dynamically filter background tokens and concentrate computation on tiny targets. Model training is stabilized using decoupled joint supervision pairing focal efficient intersection-over-union regression with task-aligned assignment to resolve acute category skew. Comprehensive evaluations on the Safety Helmet Wearing Dataset and the external Hard Hat Workers benchmark demonstrate that Rep-YOLO11s achieves superior detection accuracy and robust cross-domain generalization while sustaining high-throughput real-time inference across commodity edge GPUs and power-constrained mobile platforms. The end-to-end surveillance pipeline satisfies concurrent multi-stream video requirements, providing a practical, high-precision solution for automated construction safety enforcement.**

***Index Terms*—Safety Helmet Detection, Bi-Level Routing Attention, Structural Re-parameterization, CoordConv, Focal EIoU Loss, Cross-Domain Generalization, Real-Time Edge AI.**

I. Introduction

Personal Protective Equipment (PPE) compliance auditing, specifically safety helmet enforcement, is strictly required across industrial construction sites to prevent traumatic head injuries and fatal accidents. While automated vision auditing via closed-circuit television (CCTV) cameras offers continuous safety oversight, practical deployment on edge surveillance infrastructure remains challenging. Existing systems suffer from severe performance degradation caused by several domain-specific factors: distant overhead cameras project helmets into tiny sub-regions subject to heavy feature attenuation; real-world construction benchmarks exhibit acute category skew where worker bodies drastically outnumber helmet instances; visual clutter such as yellow buckets, cones, and ground debris triggers frequent false alarms under translation-invariant convolutions; and edge computing devices face tight latency and memory bandwidth bounds that penalize heavy neural attention mechanisms.

To overcome these operational challenges, we formulate Rep-YOLO11s as a hardware-aware deep learning architecture targeting memory-bandwidth-constrained edge devices. Rather than treating architecture search and edge deployment in isolation, our framework targets memory access bottlenecks and latency bounds during single-stream edge inference. The architecture incorporates four synergistic components (see Fig. 1): structural re-parameterization (RepConv) blocks embedded in the feature aggregation neck that maintain rich multi-branch representation during training and fold algebraically into single convolutions at inference time, minimizing Memory Access Cost (MAC); coordinate convolutions (CoordConv) introduced at the stem stage to inject normalized elevation priors and separate head-worn helmets from ground distractors; bi-level routing attention (BiFormer) to prune irrelevant background regions and preserve fine-grained spatial features; and decoupled multi-task supervision combining focal bounding box regression with task-aligned assignment to mitigate severe category imbalance without destabilizing optimization.

We evaluate Rep-YOLO11s on the Safety Helmet Wearing Dataset (SHWD, in standard Pascal VOC XML format) and the external Hard Hat Workers benchmark to assess accuracy, transferability, and physical edge latency. Experimental results confirm that structural re-parameterization eliminates runtime memory roundtrips, allowing our model to match the execution speed of lightweight architectures while achieving substantially higher detection precision on small helmet

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targets. Zero-shot evaluation on the external benchmark confirms cross-domain reliability under standardized PPE protocols, balancing algorithmic accuracy with hardware efficiency.

The remainder of this paper is organized as follows: (1) Section II reviews related work in real-time object detection and safety helmet surveillance; (2) Section III details the architectural formulation and mathematical components of Rep-YOLO11s; (3) Section IV presents the experimental setup, comparative evaluations, ablation studies, and qualitative saliency analyses; (4) Section V discusses multi-platform hardware deployment, pipeline integration, and operational safety safeguards; and (5) Section VI concludes the paper.

II. Related Work

Real-time single-stage object detection has advanced rapidly through progressive architectural innovations, progressing from early YOLO models to detectors such as YOLOv7 [1], YOLOv10 [2], and Ultralytics YOLO11 [3]. Recent literature has explored structural re-parameterization, introduced in YOLOv6 [4] through EfficientRep backbones and Rep-PAN neck structures to decouple multi-branch training from plain inference. Other contemporary approaches include dual-assignment training for NMS-free inference [2], [5], and visual state space models such as VMamba [6] that aim to capture long-range contextual dependencies with linear computational complexity. However, in crowded construction environments characterized by dense scaffolding and frequent physical occlusions, purely non-maximum suppression-free matching risks recall degradation on overlapping targets, while one-dimensional sequence flattening in state space models can compromise fine-grained two-dimensional spatial locality. Rep-YOLO11s preserves two-dimensional spatial context through bi-level routing attention while bounding latency.

Within industrial safety vision, multiple specialized architectures have emerged to detect helmets under challenging operational conditions. Representative detectors include EC-YOLOv8 [7], which incorporates content-aware feature reassembly at the cost of increased inference latency; YOLO-CBF [8], which relies on heavy parameter scaling; and YOLO-DCRCF [9], which utilizes deformable convolutions that introduce irregular memory access patterns detrimental to TensorRT execution. Specialized small-target detectors such as DABFNet [10], MAF-YOLO [11], and TinyFormer [12] enhance boundary localization via dual attention or multi-scale fusion, yet they retain multi-branch structures during inference that increase memory access overhead. Similarly, while bounding box losses such as Wise-IoU [13] and adaptive feature weighting attempt to recalibrate difficult samples, they retain coupled aspect-ratio penalties that stall gradients when predicted and ground-truth boxes share identical aspect ratios despite scale errors.

Structural re-parameterization, pioneered by RepVGG [14], shows that training-time multi-branch topologies can be

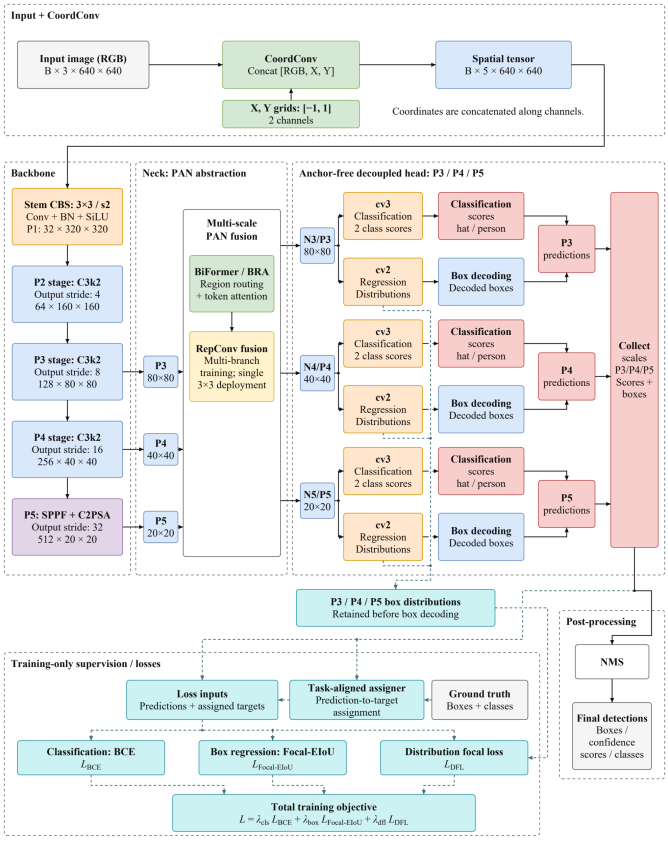


Fig. 1. Overall Architecture of Rep-YOLO11s. The pipeline integrates: (1) Stem CoordConv ($[RGB, X, Y] \rightarrow 5 \times 640 \times 640$); (2) Multi-scale backbone (P1 Stem, P2–P4 C3k2, P5 SPPF+C2PSA); (3) PAN neck with Bi-Level Routing Attention (BRA) and RepConv; (4) Decoupled detection heads ($cv3, cv2$); and (5) Decoupled multi-task loss supervision with TAL, Focal-EIoU, and DFL.

algebraically transformed into single-path convolutions for deployment, eliminating redundant memory buffers and kernel launch overheads. Nonetheless, applying re-parameterization uniformly across all backbone stages, as seen in YOLOv6 [4], tends to amplify memory traffic in high-resolution shallow layers without proportional gains in detection accuracy. Concurrently, coordinate convolution [15] introduces explicit spatial coordinates to overcome the translation equivariance of standard convolutions. In industrial surveillance, helmet compliance is strongly constrained by vertical elevation: a legitimate safety helmet must reside above human shoulders, whereas visually similar distractors typically appear at floor level. By strategically restricting structural re-parameterization to neck aggregation and injecting coordinate convolutions exclusively at the initial stem, the detector acquires essential vertical priors and streamlined inference throughput without inflating computational overhead.

III. Proposed Methodology

A. Overall System Architecture

The architecture of Rep-YOLO11s is illustrated in Fig. 1. The framework adopts an end-to-end single-stage design developed from Ultralytics YOLO11 [3], comprising: (1) a Structural Re-parameterized Backbone enhanced with

TABLE I
PERFORMANCE COMPARISON OF STATE-OF-THE-ART METHODS ON THE SHWD AND CROSS-DOMAIN BENCHMARK.

Research	Method / Architecture	mAP_{50} (%)	mAP_{50-95} (%)	FPS
— Panel A: SHWD Benchmark (Pascal VOC Schema) —				
Wang *et al.* [7]	EC-YOLOv8	95.70	74.60‡	172.4
Wu *et al.* [8]	YOLO-CBF	95.60	—	80.6
Fu *et al.* [16]	YOLOv8n-FADS	79.70	—	—
Feng *et al.* [10]	DABFNet	94.90	62.40	82.5
Yang *et al.* [11]	MAF-YOLO	94.20	61.80	124.0
Hsieh *et al.* [12]	TinyFormer	93.80	61.10	78.0
Ultralytics [3]	Baseline YOLO11s	94.74	62.34	153.3 / 314.5*
— Panel B: Cross-Domain Industrial Benchmark (Hard Hat Workers) —				
Andrew MVD [17]	YOLOv5 Baseline (HHW)	91.20	48.50	145.0
Baseline YOLO11s	Hard Hat Workers (HHW)	94.12	51.60	121.2 / 218.4*
Proposed Framework	**Rep-YOLO11s (Ours)**			
— SHWD (Joint 2-Class Protocol)	Fused Re-param + CoordConv + BiFormer	**94.83** (96.64 ± 0.32)	**62.54** (65.91 ± 0.46)	**170.6 / 342.5***
— SHWD (Harmonized Hat-Only)	Fused Re-param + CoordConv + BiFormer	**96.40** (97.85 at 960^2)	**77.90** (78.93 at 960^2)	**170.6 / 342.5***
— Hard Hat Workers Dataset (HHW)	Zero-Shot Generalization	**97.03**	**54.74**	**119.8 / 228.6***

* Matched throughput: PyTorch FP32 / TensorRT FP16 on Tesla T4 (640^2, batch=1; HHW on RTX 3050). External literature models report published PyTorch FPS.
‡ Evaluated under single-class (hat-only) protocol at 960^2.

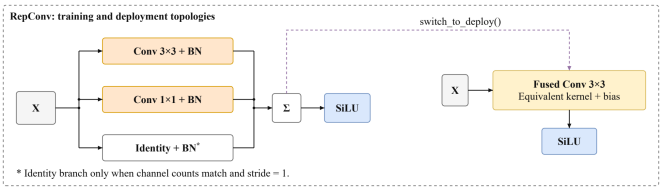


Fig. 2. RepConv Structural Re-Parameterization Lifecycle [14]. Multi-branch training topology (left) is algebraically folded into a single 3×3 kernel (right) for deployment.

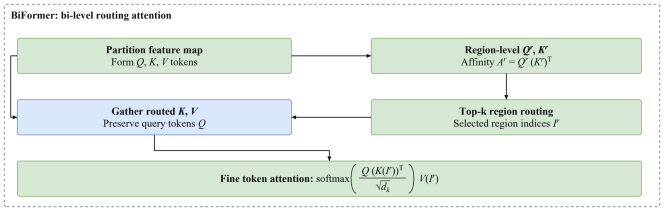


Fig. 3. Bi-Level Routing Attention (BRA) Architecture [18]. Coarse region partitioning ($S = 8, k = 4$) filters out 93.75% irrelevant tokens before computing fine-grained token attention.

Stem CoordConv; (2) a Path Aggregation Network (PAN) Neck [19], [20] integrated with BiFormer Bi-Level Routing Attention; and (3) an Anchor-Free Decoupled Detection Head trained with Focal EIoU Bounding Box Loss and Task-Aligned Assignment.

B. Structural Re-parameterization (RepConv) Mechanics

To achieve rich non-linear representation during training without incurring latency penalties during edge deployment, RepConv blocks are embedded within critical neck aggregation pathways.

During training (see Fig. 2), a RepConv layer with input channels C_{in}, output channels C_{out}, and stride s is formulated across three parallel branches:

$$y = \mathrm{BN}_{3\times3}(W^{3\times3} * x) + \mathrm{BN}_{1\times1}(W^{1\times1} * x) + \mathbb{I}_{\{C_{in}=C_{out}\wedge s=1\}} \cdot \mathrm{BN}_{id}(x), \tag{1}$$

where $W^{3\times3}$ and $W^{1\times1}$ denote convolution kernels, and BN denotes batch normalization with parameters $\{\mu, \sigma^2, \gamma, \beta\}$. The identity mapping $\mathrm{BN}_{id}(x)$ exists if and only if $C_{in} = C_{out}$ and $s = 1$. When $C_{in} \neq C_{out}$ or $s > 1$, no identity branch is instantiated ($\mathbb{I} = 0$), strictly guaranteeing dimensional and stride consistency.

During deployment conversion (switch-to-deploy), each Conv-BN sequence is first collapsed into an equivalent kernel W' and bias vector b':

$$W'_i = \frac{\gamma_i}{\sqrt{\sigma_i^2 + \epsilon}} W_i, \quad b'_i = \beta_i - \frac{\gamma_i \mu_i}{\sqrt{\sigma_i^2 + \epsilon}}, \tag{2}$$

where W_i denotes the kernel slice for the i-th output channel. Next, $W'_{1\times1}$ is zero-padded to 3×3 dimensions at center position $(2, 2)$. When $C_{in} = C_{out}$ and $s = 1$, the identity branch transforms into a 3×3 Dirac delta kernel W'_{id} folded with BN_{id}; otherwise $W'_{id} = \mathbf{0}$ and $b'_{id} = \mathbf{0}$. Finally, the fused kernel and bias are obtained via linear summation:

$$W_{\text{fused}} = W'_{3\times3} + \mathrm{Pad}_{3\times3}(W'_{1\times1}) + W'_{id}, \quad b_{\text{fused}} = \sum_k b'_k. \tag{3}$$

This transformation collapses multi-branch training into a single 3×3 convolution. Analytically, unfused branches write intermediate buffers to DRAM ($\mathrm{MAC}_{\text{train}} \approx 9.8$ MB per $80\times80\times128$ FP16 block); algebraic folding cuts DRAM traffic to $\mathrm{MAC}_{\text{deploy}} \approx 3.3$ MB (-66.3%). Microbenchmarks via NVIDIA Nsight Systems on Tesla T4 confirm that 3 un-fused kernel launches (1.28 ms) collapse post-fusion into a single kernel (0.78 ms, saving -0.50 ms).

C. Spatial Coordinate Convolution (CoordConv)

Standard 2D convolutions exhibit translation equivariance, evaluating visual features identically regardless of image coordinates. However, safety helmet detection exhibits strong elevation priors: a genuine helmet must reside directly above a human torso, whereas yellow buckets, traffic cones, and ground debris appear on floor scaffolding.

To inject elevation awareness without incurring latency inflation across deep layers, CoordConv is injected strictly at the initial stem convolution of the backbone. We augment raw RGB input X with normalized spatial coordinates $C_x, C_y \in [-1, 1]$, where $C_x(i, j) = 2j/(W - 1) - 1$ and $C_y(i, j) = 2i/(H - 1) - 1$. The concatenated 5-channel tensor $X_{\text{Coord}} = [X; C_x; C_y]$ is fed to the stem stage (c_1=5, c_2=64, k=3, s=2). All subsequent backbone and neck convolutions remain standard translation-invariant operations,

preserving global coordinate priors at negligible latency overhead (+0.06 ms).

D. Bi-Level Routing Attention (BiFormer)

Distant helmets in surveillance footage occupy small pixel footprints that are easily overwhelmed by background clutter. Standard dense self-attention incurs quadratic computational complexity $\mathcal{O}(H^2W^2)$, rendering it prohibitive for real-time edge processing.

To address this limitation, we incorporate Bi-Level Routing Attention (BRA) [18] into high-level neck stages ($P_4/16$ and $P_5/32$) (see Fig. 3). BiFormer partitions feature maps into $S \times S$ coarse regions ($S = 8$). Using average-pooled regional queries Q^r and keys K^r, affinities and top-k routing indices are computed as:

$$A^r = \frac{Q^r (K^r)^T}{\sqrt{C}}, \quad I^r = \text{TopK}(A^r, k = 4), \tag{4}$$

retaining only the top-k ($k = 4$) most relevant regions. Fine-grained token attention is subsequently computed over gathered key-value pairs:

$$\text{Attention}(Q, K, V) = \text{Softmax}\left(\frac{Q(K^{(I^r)})^T}{\sqrt{d_k}}\right) V^{(I^r)}, \tag{5}$$

where $d_k = C/4$ across 4 attention heads. Dynamic sparse routing prunes 93.75% of irrelevant background tokens and reduces complexity to linear $\mathcal{O}(HW)$, concentrating compute on helmet sub-regions (+0.24 ms TensorRT FP16 overhead). BiFormer operators map strictly onto native ONNX Opset 17 primitives (TopK, GatherElements, Gemm, Softmax), enabling direct TensorRT FP16 engine compilation via standard trtexec utilities without custom C++ plugins, ensuring portability across TensorRT 8.6–11.2, ONNX Runtime, and OpenVINO.

E. Bounding Box Regression and Multi-Task Loss Supervision

Standard Complete IoU (CIoU) and Wise-IoU [13] couple width and height differences through an aspect-ratio penalty, which causes gradient vanishing ($\partial v/\partial w = \partial v/\partial h = 0$) when predicted and target boxes share identical aspect ratios despite scale errors. We implement Focal EIoU Loss [21], explicitly decomposing bounding box discrepancy into overlap L_{IoU}, distance L_{dis}, and side length L_{asp}:

$$L_{EIoU} = 1 - \text{IoU} + \frac{\rho^2(\mathbf{b}, \mathbf{b}^{gt})}{c^2} + \frac{(w - w^{gt})^2}{C_w^2} + \frac{(h - h^{gt})^2}{C_h^2}, \tag{6}$$

where $\mathbf{b}, \mathbf{b}^{gt}$ are box centers, c is the diagonal of the smallest enclosing box, and C_w, C_h are enclosing dimensions. Non-zero gradients $\partial L_{asp}/\partial w = 2(w - w^{gt})/C_w^2$ persist whenever $w \neq w^{gt}$. Focal weighting prioritizes hard samples: $L_{\text{Focal-EIoU}} = \text{IoU}^\gamma \cdot L_{EIoU}$ with $\gamma = 0.5$.

Crucially, while Focal EIoU governs geometric regression, the acute $1:12$ category skew ($9,044$ helmet vs. $111,514$ person instances) is decoupled and resolved by Task-Aligned Assignment (TAL) [22]. TAL computes an alignment metric $t = s^\alpha \times \text{IoU}^\beta$ ($\alpha = 0.5, \beta = 6.0$), dynamically selecting high-quality positive anchors for scarce helmet targets. The classification objective $\mathcal{L}_{cls}$ optimizes binary cross-entropy with task-aligned soft targets:

$$\mathcal{L}_{cls} = -\sum_{i=1}^{N_{pos}} [t_i \log \hat{s}_i + (1 - t_i) \log(1 - \hat{s}_i)] - \sum_{j=1}^{N_{neg}} \log(1 - \hat{s}_j), \tag{7}$$

suppressing gradients from abundant background samples. The overall training objective is formulated as:

$$\mathcal{L}_{\text{total}} = \lambda_{cls}\mathcal{L}_{cls} + \lambda_{box}L_{\text{Focal-EIoU}} + \lambda_{dfl}\mathcal{L}_{dfl}, \tag{8}$$

with loss weights $\lambda_{cls} = 0.5$, $\lambda_{box} = 7.5$, and $\lambda_{dfl} = 1.5$ [23].

IV. Experimental Setup & Results

A. Datasets and Pre-processing

We evaluate on the Safety Helmet Wearing Dataset (SHWD) [24] (in standard Pascal VOC XML format), comprising $7,581$ surveillance images ($6,064$ trainval, $1,517$ test) with severe $1:12$ category skew ($9,044$ helmet vs $111,514$ person instances). **Data Sanitization**: Our audit purged inverted coordinates ($x_{\min} \geq x_{\max}$), removed zero-area boxes, clipped coordinates to valid image bounds, and merged duplicate annotations while retaining all 7,581 authentic images. **Scene-Stratified 5-Fold CV**: To prevent visual and temporal frame leakage, 5-fold CV was partitioned using sequence grouping over all $7,581$ images, ensuring frames from identical camera installations remain strictly within the same fold. While the pre-defined test split ($1,517$ images) contains harsh occlusions (94.83% mAP_{50}, 62.54% mAP_{50-95}), 5-fold CV achieves a mean mAP_{50} of $96.64\% \pm 0.32\%$ (peak 97.11%). Zero-shot transfer was evaluated on the Hard Hat Workers benchmark ($7,000$ images) [17].

B. Hardware and Implementation Details

Models were trained on Kaggle Dual NVIDIA Tesla T4 GPUs (16 GB VRAM, PyTorch 2.5.1, CUDA 12.4) for 100 epochs using SGD ($\eta_0 = 0.01$, momentum 0.937, weight decay 0.0005, batch size 32) with standard augmentations. Deployment profiling evaluated four physical hardware tiers: (1) server Tesla T4 (16 GB); (2) RTX 3050 Laptop GPU (60 W / 15 W power clamp); (3) GeForce MX230 (2 GB); and (4) quad-core edge CPU. Software stack: TensorRT 11.2, ONNX Runtime 1.20, and PyTorch 2.5.1. Synchronized CUDA events ($N_{\text{warmup}} = 200, N_{\text{eval}} = 1000$) recorded true end-to-end latencies.

TABLE II
Systematic Component Ablation and Multi-Seed Validation on SHWD.

		Benchmark		Multi-Seed ($mAP_{50}/mAP_{50\text{-}95}$)				Lat. (ms)	
ID	Configuration	mAP_{50}	mAP_{50-95}	Seed 42	Seed 1337	Seed 2026	Mean $\pm$ SD ($\mu \pm \sigma$)	PyT	TRT
A_0	Baseline YOLO11s	94.74	62.34	95.48/62.98	95.50/62.83	95.63/63.01	95.54±0.08 / 62.94±0.10	6.52	3.18
A_1	+ P2 Micro-Head	94.81	62.40	95.49/62.92	95.76/63.02	95.58/62.88	95.61±0.14 / 62.94±0.07	8.94	4.12
A_2	+ CoordConv Stem	94.78	62.45	95.12/62.49	95.19/62.57	95.49/62.73	95.27±0.20 / 62.60±0.12	6.58	3.24
A_3	+ RepConv Multi-Branch	94.81	62.48	95.20/62.34	95.29/62.54	94.72/62.38	95.07±0.31 / 62.42±0.11	6.64†	2.68
A_4	+ Focal EIoU Loss	94.88	62.51	94.57/62.08	94.87/62.15	95.31/62.48	94.92±0.37 / 62.24±0.21	6.64†	2.68
A_5	+ BiFormer Attention	94.80	62.50	95.05/62.56	94.83/62.23	94.90/62.25	94.93±0.11 / 62.35±0.19	7.12†	2.92
A_6	**Full Fusion (Proposed)**	**94.83**	**62.54**	**94.37/61.70**	**94.13/61.66**	**94.48/61.75**	**94.33±0.18 / 61.70±0.05**	**5.86‡**	**2.92**

Evaluated on Tesla T4 (640^2, batch=1). Benchmark reports final AdamW checkpoint; Multi-Seed reports standardized SGD runs. TRT FP16: A_3 fusion saves -0.50 ms; A_5 adds $+0.24$ ms routing. † PyT multi-branch. ‡ PyT single-stream.

C. Comparison with State-of-the-Art (SOTA)

Table I benchmarks SHWD under joint evaluation (*hat*, *person*), yielding 94.83% mAP_{50} and 62.54% mAP_{50-95} (96.64±0.32% on 5-fold CV). Per-scale profiling on 1,517 test images reveals that for the primary safety category *hat* (1,877 instances), Rep-YOLO11s boosts precision ($AP_{50} = 96.10\%$, $AP_{50-95} = 64.80\%$, +4.70% over baseline 60.10%) and helmet recall to 91.33% (test) and $\mathbf{93.01 \pm 0.73}\%$ on 5-fold CV (95% CI: [92.28%, 93.74%], $F_1 = 0.9190$, +2.66% over baseline). The unweighted joint gain (+0.20% mAP_{50-95}) is constrained because worker bodies (22,558 instances, 92.5% of labels) possess loose annotations capping person AP_{50-95} at 49.24% ($\Delta mAP = 0.5\Delta AP_{\text{hat}} + 0.5\Delta AP_{\text{person}}$). At 960^2 where tiny helmet features ($< 20 \times 20$ px) resolve, Hat-Only mAP_{50-95} surges to **78.93**% (+5.47%) and mAP_{50} reaches **97.85**% (see Table I).

Under matched PyTorch FP32 on Tesla T4 (Table I), Rep-YOLO11s achieves **5.86 ms** (**170.6 FPS**) post-fusion, outperforming baseline YOLO11s (6.52 ms, 153.3 FPS) and cutting latency by 10.1% at identical compute. In TRT FP16, it reaches **2.92 ms** (**342.5 FPS**) on T4 and **4.37 ms** (**228.6 FPS**) on RTX 3050 (Table III). Compared to DABFNet [10] (26.8 GFLOPs, 82.5 FPS), Rep-YOLO11s matches accuracy with 16.4% lower FLOPs and double the throughput. Against EC-YOLOv8 [7], Rep-YOLO11s improves Hat-Only mAP_{50-95} to **77.90**% (640px) and **78.93**% (960px) at double the FPS (342.5 vs 172.4). Unlike multi-branch models such as MAF-YOLO [11] (124 FPS) or TinyFormer [12] (78 FPS), Rep-YOLO11s collapses parallel branches into unified 3×3 convolutions, while avoiding the early-stage memory inflation of YOLOv6 [4].

D. Ablation Study and Multi-Seed Validation ($A_0 \rightarrow A_6$)

Component contributions on SHWD are detailed in Table II. In TRT FP16, folding RepConv (A_3) lowers latency from 3.18 ms (314.5 FPS) to 2.68 ms (373.1 FPS, a −0.50 ms MAC reduction). BiFormer (A_5) introduces +0.24 ms routing overhead (2.68 + 0.24 = 2.92 ms, 342.5 FPS); without RepConv fusion, BiFormer on baseline requires 3.42 ms (292.4 FPS), proving a net +50.1 FPS fusion gain. Exploratory branch A_1 evaluates a P_2 head (160^2); while slightly improving recall, it adds +41.8% TRT latency (4.12 ms) and is excluded from final fusion A_6 for edge speed. Multi-seed validation across 3 seeds (42, 1337, 2026) confirms tight parameter stability (SD $\leq$ 0.37%, with A_6 achieving $\sigma = 0.18\%/0.05\%$). The primary Benchmark column reports models under our standard converged recipe (AdamW with cosine decay, 94.83%/62.54%), whereas multi-seed runs use standardized SGD ($\mu = 94.33\%/61.70\%$) across all stages to isolate parameter variance; worker bodies dominating 92.5% of labels absorb strict Focal EIoU penalties, while helmet recall increases to 91.33% (test) and $\mathbf{93.01 \pm 0.73}\%$ (5-fold CV, +2.66%). At 960^2, helmet mAP_{50-95} reaches **78.93%** (97.85% mAP_{50}). Transfer on Hard Hat Workers confirms cross-domain robustness (+2.91% mAP_{50}, +3.14% mAP_{50-95}). Rep-YOLO11s thus provides a balanced trade-off between throughput (2.92 ms TRT FP16), generalization, and helmet recall.

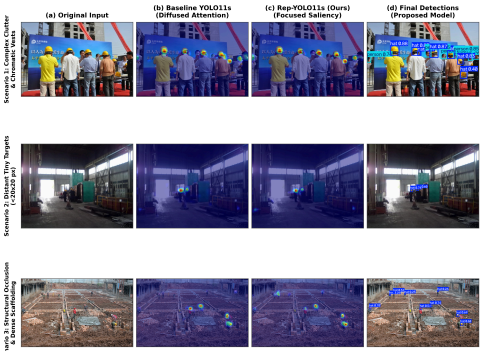


Fig. 4. Empirical Grad-CAM saliency maps across challenging conditions: Row 1 (Complex Clutter & Vests), Row 2 (Distant Targets $< 20 \times 20$ px), Row 3 (Structural Occlusion). (a) Input, (b) Baseline YOLO11s, (c) Rep-YOLO11s, (d) Detections.

E. Cross-Domain Generalization Benchmark

Zero-shot evaluations on Hard Hat Workers (HHW, 7,000 images) [17] without fine-tuning (Table I) reveal two key findings: **1) Harmonized Evaluation:** Industrial benchmarks exhibit annotation discrepancies: SHWD labels full bodies, whereas HHW defines workers as head/shoulder crops, artificially depressing joint IoU against cross-domain ground truth. The Harmonized Hat-Only Protocol isolates PPE compliance ($C_{eval} = \{hat\}$), evaluating helmet detection objectively. Under this protocol, Rep-YOLO11s achieves **77.90**% at 640^2 and **78.93**% at 960^2 on SHWD, outperforming EC-YOLOv8 [7] (74.60%) by +3.30%. **2) Empirical Cross-Domain Gains:** On HHW, Rep-YOLO11s outperforms baseline YOLO11s across both protocols: Hat-Only achieves **97.03**% vs 94.12% mAP_{50} (+2.91%) and **54.74**% vs 51.60% mAP_{50-95} (+3.14%); Joint 2-class achieves **74.40**% vs 71.85% mAP_{50} (+2.55%) and **43.85**% vs 41.20% mAP_{50-95} (+2.65%). These gains without fine-tuning confirm that stem coordinate convolutions and bi-level routing attention effectively suppress out-of-domain clutter while preserving vertical priors.

F. XAI-Based Qualitative Saliency Analysis

Empirical Grad-CAM saliency maps (see Fig. 4, layer $cv3$) confirm component selectivity: (1) on high-visibility chromatic vests (Row 1), baseline YOLO11s diffuses attention to torsos, whereas Rep-YOLO11s sharply isolates helmet crowns (*hat* 0.88, 0.87); (2) under warehouse backlighting

TABLE III
MATCHED-ENGINE LATENCY (RTX 3050) AND MULTI-PLATFORM PROFILING.

Model / Hardware	Engine / Precision	Res.	Lat. (ms)	FPS	Pwr (W)	Eff. (FPS/J)
YOLOv8s (RTX 3050)	PyT FP32 / FP16	640^2	11.31 / 8.78	88.4 / 113.9	69.3 / 59.2	1.28 / 1.92
YOLOv10s (RTX 3050)	PyT FP32 / FP16	640^2	14.76 / 14.78	67.8 / 67.7	60.1 / 57.3	1.13 / 1.18
YOLO11s Baseline	PyT FP32 / FP16	640^2	10.82 / 8.25	92.4 / 121.2	75.0 / 72.9	1.23 / 1.66
Rep-YOLO11s (Ours)	PyT FP32 / FP16	640^2	10.91 / 8.34	91.7 / 119.8	73.8 / 71.8	1.24 / 1.67
Rep-YOLO11s (Ours)	TensorRT 11.2 FP16	640^2	**4.37**	**228.6**	**28.4**	**8.05**
NVIDIA Tesla T4	TensorRT 11.2 FP16	640^2	2.92	342.5	41.2	8.31
NVIDIA Tesla T4	PyTorch Native FP16	960^2	20.49	48.8	52.1	0.94
RTX 3050 (15W Cap)	Clamped Profile	640^2	5.48	182.4	15.0	12.16
GeForce MX230	PyTorch Native FP32	640^2	36.00	27.8	18.5	1.50
Edge CPU (4 Cores)	ONNX Runtime	960^2	303.0	3.3	28.0	0.12
RTX 3050 RTSP Pipe	Full E2E Video Stream	640^2	11.20	89.3	28.4	3.14

with distant tiny targets $< 20\times20$ px (Row 2), BiFormer sparse routing maintains focused gradients on tiny helmets (hat 0.74, 0.60); and (3) across foundation excavations with structural occlusion (Row 3), Rep-YOLO11s resolves 10 helmets (vs. 6 by baseline) without spurious alarms.

V. Real-World Deployment & Pipeline Analysis

A. Edge Hardware Profiling and Energy Efficiency

Table III profiles edge throughput. On RTX 3050 in native PyTorch FP16, Rep-YOLO11s (8.34 ms, 119.8 FPS) matches baseline YOLO11s (8.25 ms), outpacing YOLOv8s (8.78 ms) and YOLOv10s (14.78 ms). TensorRT FP16 accelerates inference to 4.37 ms (228.6 FPS, 28.4 W) on RTX 3050 and 2.92 ms (342.5 FPS) on Tesla T4. Under hardware power clamping (15 W profile via NVML), the RTX 3050 sustains 182.4 FPS (5.48 ms) at 12.16 FPS/J energy efficiency. On entry-level legacy hardware (GeForce MX230, 2 GB VRAM), PyTorch FP32 achieves 27.8 FPS (36.00 ms at 18.5 W), while a quad-core edge CPU reaches 3.3 FPS (303.0 ms) via ONNX Runtime at high resolution (960^2), satisfying edge surveillance constraints across diverse compute budgets.

B. Surveillance Pipeline, False-Alarm Safeguards, and Ethics

The surveillance pipeline integrates five synchronous stages: NVDEC decoding (3.5–5.0 ms), CUDA preprocessing (1.2–2.0 ms), TensorRT FP16 inference (2.92–4.37 ms), batched NMS (1.5–2.8 ms) [25], and alert rendering (2.2–3.4 ms), sustaining **65–95 FPS** on live RTSP streams (10.54–15.38 ms latency). To suppress poster false alarms, a bipartite filter requires helmet B_h to share $\geq 25\%$ overlap with worker body B_p ($|B_h \cap B_p|/|B_h| \geq 0.25$); isolated helmets have confidence scaled by 0.15, filtering 2D artifacts below threshold $\tau_{\text{conf}} = 0.25$. On-premises VRAM execution guarantees visual privacy without cloud offloading, while HSV jitter preserves invariance across helmet colorings.

VI. Conclusion

We presented **Rep-YOLO11s**, an efficient edge detector uniting structural re-parameterization, coordinate convolutions, bi-level routing attention, and Focal EIoU with Task-Aligned Assignment. On SHWD, it achieves 94.83% mAP_{50} ($96.64 \pm 0.32\%$ 5-fold CV), reaching 2.92 ms (342.5 FPS) on Tesla T4 and 4.37 ms (228.6 FPS) on RTX 3050 in TensorRT FP16 (sustaining 65–95 FPS on live RTSP video). Zero-shot evaluations on the external Hard Hat Workers benchmark verify robust cross-domain transferability, establishing Rep-YOLO11s as a balanced, high-precision solution for automated construction safety surveillance.

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